

VPA A-Level Computer Science Bilingual Glossary — 150+ Terms

Victoria Park Academy — A-Level Computer Science

Bilingual Glossary: 150+ Terms

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Format: English | 中文 | Pinyin | Definition

Note: Terms marked with □ are A-Level specific (not typically covered at IGCSE level).

Unit 1: Data Representation

English	中文	Pinyin	Definition
Binary	二进制	□èrjìnzhì	A number system using base 2, with digits 0 and 1.
Denary / Decimal	十进制	Shíjìnzhì	A number system using base 10, with digits 0–9.
Hexadecimal	十六进制	Shíliù jìnzhì	A number system using base 16, with digits 0–9 and A–F.
Bit	位	Wèi	The smallest unit of data, representing 0 or 1.
Byte	字节	Zìjié	A group of 8 bits, the standard unit of storage.
Nibble	半字节	Bàn zìjié	A group of 4 bits, half of a byte.
Two's Complement	补码	Bǔmǎ	A method for representing signed integers in binary, allowing subtraction by addition.

English	中文	Pinyin	Definition
Sign and Magnitude	符号幅度	Fúhào fúdù	A method for representing signed integers where the first bit is the sign and the rest represent the magnitude.
Overflow	溢出	Yìchū	An error that occurs when a calculation produces a result too large to be stored.
Underflow	下溢	Xià yì	An error when a number is too small (close to zero) to be represented.
☐ Floating-Point	浮点数	Fúdiǎn shù	A representation of real numbers using a mantissa and an exponent, allowing a wide range of values.
☐ Mantissa	尾数	Wěishù	The fractional part of a floating-point number, representing the significant digits.
☐ Exponent	指数	Zhǐshù	The power to which the base is raised in a floating-point number.
☐ Normalisation	规范化	Guīfàn huà	The process of adjusting a floating-point number so the mantissa is within a standard range (e.g., $0.5 \leq \text{mantissa} < 1$).
ASCII	美国信息交换标准代码	Měiguó xìnxī jiāohuàn biāozhǔn dàimǎ	A 7-bit character encoding standard representing 128 characters.
Unicode	统一码	Tǒngyī mǎ	A universal character encoding standard supporting multiple writing systems, including Chinese.

English	中文	Pinyin	Definition
Binary-Coded Decimal (BCD)	二进制编码的十进制	èrjìnzhì biānmǎ de shíjìnzhì	A representation where each decimal digit is stored as a 4-bit binary number.
Gray Code	格雷码	Géléi mǎ	A binary code where consecutive values differ by only one bit, reducing errors in digital systems.
Resolution	分辨率	Fēnjiàn lǜ	The number of pixels per unit of length in an image, or the number of bits used to represent a sample in sound.
Colour Depth	色深	Sè shēn	The number of bits used to represent the colour of a single pixel.
Metadata	元数据	Yuán shùjù	Data that describes other data, such as image dimensions, file type, or creation date.
Sampling Rate	采样率	Cǎiyàng lǜ	The number of samples taken per second when digitising sound.
Sampling Resolution	采样分辨率	Cǎiyàng fēnjiàn lǜ	The number of bits used to represent each sample of sound.
□ Nyquist Theorem	奈奎斯特定理	Nài kuí sī tè dìnglǐ	States that the sampling rate must be at least twice the highest frequency in the signal to avoid loss.
Bit Rate	比特率	Bǐtè lǜ	The number of bits processed per unit of time, typically bits per second (bps).
Lossless Compression	无损压缩	Wúsuǎn yāsuō	Compression that reduces file size without losing any data, allowing perfect reconstruction.

English	中文	Pinyin	Definition
Lossy Compression	有损压缩	Yǒusǔn yāsuō	Compression that reduces file size by permanently removing some data, sacrificing quality.
Run-Length Encoding (RLE)	行程长度编码	Xíngchéng chángdù biānmǎ	A simple lossless compression method replacing repeated values with a count and value.

Unit 2: Communication

English	中文	Pinyin	Definition
Protocol	协议	Xiéyì	A set of rules governing data communication between devices.
TCP/IP	传输控制协议/网际协议	Chuánshū kòngzhì xiéyì / Wǎngjì xiéyì	The fundamental suite of protocols for internet communication.
HTTP	超文本传输协议	Chāo wénběn chuánshū xiéyì	The protocol used for transferring web pages on the internet.
HTTPS	安全超文本传输协议	Ānquán chāo wénběn chuánshū xiéyì	HTTP with encryption (TLS/SSL) for secure communication.
FTP	文件传输协议	Wénjiàn chuánshū xiéyì	A protocol for transferring files between computers on a network.
SMTP	简单邮件传输协议	Jiǎndān yóujiàn chuánshū xiéyì	The protocol for sending email messages between servers.
Packet Switching	分组交换	Fēnzǔ jiāohuàn	A method where data is broken into packets that travel independently and are reassembled at the destination.
Circuit Switching	电路交换	Diànlù jiāohuàn	A method where a dedicated communication path is established before data transfer.

English	中文	Pinyin	Definition
Router	路由器	Lùyóu qì	A device that forwards data packets between networks based on IP addresses.
Switch	交换机	Jiāohuàn jī	A device that connects devices within a local network and forwards data based on MAC addresses.
Bandwidth	带宽	Dàikuān	The maximum rate of data transfer across a network path, measured in bits per second.
Latency	延迟	Yáncí	The time delay between sending and receiving data across a network.
Client-Server	客户端-服务器	Kèhù duān - Fúwù qì	A network architecture where clients request services from central servers.
Peer-to-Peer (P2P)	点对点	Diǎn duì diǎn	A network architecture where all devices have equal status and can act as both client and server.
Thin Client	瘦客户端	Shòu kèhù duān	A low-specification device that relies on a server for processing and storage.
Thick Client	胖客户端	Pàng kèhù duān	A device with significant local processing and storage capability.
☐ Routing Algorithm	路由算法	Lùyóu suànfǎ	An algorithm that determines the best path for data to travel across a network.
☐ Dijkstra's Algorithm	迪杰斯特拉算法	Dí jié sī tè lǎ suànfǎ	A shortest-path algorithm used in network routing to find the most efficient route.

Unit 3: Hardware

English	中文	Pinyin	Definition
Von Neumann Architecture	冯·诺依曼架构	Féng nuò yī màn jiàgòu	A computer design where instructions and data are stored in the same memory and processed sequentially.
Fetch-Decode-Execute Cycle	取指-译码-执行周期	Qǔ zhǐ - Yì mǎ - Zhíxíng zhōuqī	The fundamental cycle of a CPU: fetch instruction from memory, decode it, execute it.
Program Counter (PC)	程序计数器	Chéngxù jìshù qì	A register that holds the address of the next instruction to be fetched.
Memory Address Register (MAR)	存储器地址寄存器	Cúncǔ qì dìzhǐ jìcún qì	A register that holds the address of the memory location to be accessed.
Memory Data Register (MDR)	存储器数据寄存器	Cúncǔ qì shùjù jìcún qì	A register that holds data being transferred to or from memory.
Accumulator (ACC)	累加器	Lěijiā qì	A register that stores intermediate results of arithmetic and logical operations.
Current Instruction Register (CIR)	当前指令寄存器	Dāngqián zhǐlìng jìcún qì	A register that holds the instruction currently being decoded and executed.
Assembly Language	汇编语言	Huìbiān yǔyán	A low-level programming language with a one-to-one correspondence with machine code instructions.
Machine Code	机器码	Jīqì mǎ	Binary instructions that a CPU can execute directly without translation.

English	中文	Pinyin	Definition
Addressing Mode	寻址模式	Xúnzhǐ móshì	The way in which the operand of an instruction is specified (e.g., immediate, direct, indirect).
Clock Speed	时钟速度	Shízhōng sùdù	The number of cycles a CPU completes per second, measured in Hertz (Hz).
Word Length	字长	Zì cháng	The number of bits processed by a CPU in a single operation.
Cache Memory	高速缓存	Gāosù huǎncún	Small, fast memory located close to the CPU that stores frequently accessed data.
Pipelining	流水线	Liúshuǐ xiàn	A technique where multiple instructions are overlapped in execution to improve CPU efficiency.
Multicore Processor	多核处理器	Duō hé chǔlǐ qì	A CPU with multiple processing cores, allowing parallel execution of tasks.
☐ Paging	分页	Fēn yè	A memory management scheme that divides physical memory into fixed-size blocks called pages.
☐ Segmentation	分段	Fēn duàn	A memory management scheme that divides memory into variable-sized segments based on logical divisions.
☐ Virtual Memory	虚拟内存	Xūnǐ nèicún	A technique that uses secondary storage to extend the apparent capacity of main memory.

English	中文	Pinyin	Definition
☐ Thrashing	抖动	Dǒudòng	A condition where excessive paging causes the system to spend more time swapping than executing.
Input Device	输入设备	Shūrù shèbèi	A hardware device used to enter data into a computer system.
Output Device	输出设备	Shūchū shèbèi	A hardware device used to present data from a computer system to the user.
Storage Device	存储设备	Cúncǔ shèbèi	A hardware device used to store data persistently.
SSD	固态硬盘	Gùtài yìngpán	A storage device using flash memory with no moving parts, faster than traditional hard drives.
HDD	机械硬盘	Jīxiè yìngpán	A storage device using spinning magnetic platters to store data.
RAID	独立磁盘冗余阵列	Dúlì cípán rǒngyú zhènlìè	A technology that combines multiple disk drives for performance, redundancy, or both.

Unit 4: Software

English	中文	Pinyin	Definition
Operating System	操作系统	Cāozuò xìtǒng	System software that manages hardware resources and provides services for applications.
Memory Management	内存管理	Nèicún guǎnlǐ	The function of an OS that controls and coordinates the use of main memory.

English	中文	Pinyin	Definition
Scheduling Algorithm	调度算法	Diàodù suànfǎ	An algorithm that determines the order in which processes are executed by the CPU.
Round Robin	时间片轮转	Shíjiān piàn lúnzhuǎn	A scheduling algorithm where each process gets a fixed time slice in turn.
First-Come First-Served (FCFS)	先来先服务	Xiān lái xiān fúwù	A scheduling algorithm that processes jobs in the order they arrive.
Shortest Job First (SJF)	短作业优先	Duǎn zuòyè yōuxiān	A scheduling algorithm that prioritises the job with the shortest execution time.
Compiler	编译器	Biānyì qì	A program that translates high-level source code into machine code in one go.
Interpreter	解释器	Jiěshì qì	A program that translates and executes high-level code line by line.
Assembler	汇编器	Huìbiān qì	A program that translates assembly language into machine code.
Lexical Analysis	词法分析	Cífǎ fēnxī	The first stage of compilation, breaking source code into tokens.
Syntax Analysis	语法分析	Yǔfǎ fēnxī	The second stage of compilation, checking tokens against grammar rules and building a parse tree.
Code Generation	代码生成	Dàimǎ shēngchéng	The stage of compilation where intermediate code is translated into target machine code.
Optimisation	优化	Yōuhuà	The stage of compilation that improves code efficiency without changing its behaviour.

English	中文	Pinyin	Definition
Utility Program	实用程序	Shíyòng chéngxù	System software that performs maintenance tasks (backup, defragmentation, compression).
Defragmentation	碎片整理	Suìpiàn zhěnglǐ	Reorganising data on a disk so files are stored in contiguous blocks, improving access speed.
Backup	备份	Bèifèn	Copying data to a secondary location to protect against loss.
Compression	压缩	Yāsuō	Reducing file size by encoding data more efficiently.
Library Program	库程序	Kù chéngxù	A collection of pre-written code modules that can be reused in different programs.
Linker	链接器	Liànjiē qì	A program that combines object code and library code to create an executable file.
Loader	加载器	Jiāzài qì	A program that loads an executable into memory and prepares it for execution.

Unit 5: Security

English	中文	Pinyin	Definition
Malware	恶意软件	ǐyì ruǎnjiàn	Software designed to harm or exploit computer systems.
Virus	病毒	Bìngdú	Malware that attaches itself to legitimate programs and spreads when executed.
Worm	蠕虫	Rúchóng	Self-replicating malware that spreads across networks without user action.

English	中文	Pinyin	Definition
Trojan	木马	Mùmǎ	Malware disguised as legitimate software that tricks users into installing it.
Spyware	间谍软件	Jiàndié ruǎnjiàn	Malware that secretly monitors user activity and collects personal information.
Ransomware	勒索软件	Lèsuǒ ruǎnjiàn	Malware that encrypts user data and demands payment for its release.
Phishing	网络钓鱼	Wǎngluò diàoyú	A social engineering attack that tricks users into revealing sensitive information via fraudulent communications.
Firewall	防火墙	Fánguǒ qiáng	A security system that monitors and controls network traffic based on predetermined rules.
Proxy Server	代理服务器	Dàilǐ fúwù qì	An intermediary server that forwards requests and can filter, cache, or monitor traffic.
Encryption	加密	Jiāmì	The process of converting data into a coded form to prevent unauthorised access.
Decryption	解密	Jiěmì	The process of converting encrypted data back into its original form.
Symmetric Encryption	对称加密	Duìchèn jiāmì	Encryption using the same key for both encryption and decryption.
Asymmetric Encryption	非对称加密	Fēi duìchèn jiāmì	Encryption using a pair of keys: a public key for encryption and a private key for decryption.

English	中文	Pinyin	Definition
Public Key	公钥	Gōng yào	A key that can be freely distributed and is used to encrypt data or verify signatures.
Private Key	私钥	Sī yào	A key that must be kept secret and is used to decrypt data or create signatures.
Digital Signature	数字签名	Shùzì qiānmíng	A cryptographic mechanism that verifies the authenticity and integrity of a message.
Digital Certificate	数字证书	Shùzì zhèngshū	An electronic document that binds a public key to an identity, issued by a Certificate Authority.
Certificate Authority (CA)	证书颁发机构	Zhèngshū bānfā jīgòu	A trusted organisation that issues and verifies digital certificates.
PKI	公钥基础设施	Gōng yào jīchǔ shèshī	Public Key Infrastructure: the framework for managing digital keys and certificates.
Authentication	认证	Rènzhèng	The process of verifying the identity of a user or device.
Authorisation	授权	Shòuquán	The process of granting permissions to an authenticated user.
Biometrics	生物识别	Shēngwù shíbié	Authentication using physical characteristics (fingerprint, iris, face, voice).
Two-Factor Authentication (2FA)	双因素认证	Shuāng yīnsù rènzhèng	Authentication requiring two different types of evidence (e.g., password + phone code).

English	中文	Pinyin	Definition
CAPTCHA	验证码	Yànzhèng mǎ	A test to distinguish humans from automated systems, often using distorted text or images.
GDPR	通用数据保护条例	Tōngyòng shùjù bǎohù tiáolì	The EU General Data Protection Regulation governing data privacy and protection.
Data Subject	数据主体	Shùjù zhǔtǐ	An individual whose personal data is being processed.
Data Controller	数据控制者	Shùjù kòngzhì zhě	The entity that determines the purposes and means of processing personal data.
Data Processor	数据处理者	Shùjù chǔlǐ zhě	The entity that processes personal data on behalf of the data controller.

Unit 6: Programming

English	中文	Pinyin	Definition
Variable	变量	Biànràng	A named storage location in memory whose value can change during program execution.
Constant	常量	Chángliàng	A named value that cannot change during program execution.
Data Type	数据类型	Shùjù lèixíng	A classification that specifies the kind of data a variable can hold.
Integer	整数	Zhěngshù	A whole number without a fractional component.
Real / Float	实数/浮点数	Shíshù / Fúdiǎn shù	A number with a fractional component.
Boolean	布尔值	Bù'ěr zhí	A data type with two possible values: True or False.

English	中文	Pinyin	Definition
Character	字符	Zìfú	A single letter, digit, or symbol.
String	字符串	Zìfú chuàn	A sequence of characters.
Array	数组	Shùzǔ	A data structure that stores multiple values of the same type in a contiguous block of memory.
Record	记录	Jìlù	A composite data type that groups related fields of different types.
Selection	选择	Xuǎnzé	A control structure that chooses between alternative paths (IF, CASE).
Iteration	迭代	Diédài	A control structure that repeats a block of code (FOR, WHILE, REPEAT).
Procedure	过程	Guòchéng	A subroutine that performs a task but does not return a value.
Function	函数	Hánshù	A subroutine that performs a task and returns a value.
Parameter	参数	Cānshù	A variable used to pass data into a subroutine.
Return Value	返回值	Fǎnhuí zhí	The value sent back by a function to the calling code.
Local Variable	局部变量	Júbù biànlìang	A variable declared inside a subroutine, accessible only within that subroutine.
Global Variable	全局变量	Quánjú biànlìang	A variable declared outside all subroutines, accessible from anywhere in the program.

English	中文	Pinyin	Definition
Pass by Value	值传递	Zhí chuándì	Passing a copy of the data to a subroutine; changes do not affect the original.
Pass by Reference	引用传递	Yīnyòng chuándì	Passing the memory address of data to a subroutine; changes affect the original.
Recursion	递归	Dìguī	A programming technique where a function calls itself to solve a problem by breaking it into smaller instances.
Base Case	基准情形	Jīzhǔn qíngxíng	The condition in a recursive function that stops the recursion.
Stack Frame	栈帧	Zhàn zhèng	A block of memory allocated on the call stack for a subroutine's local variables and parameters.
File Handling	文件处理	Wénjiàn chǔlǐ	Operations to read from and write to files (open, read, write, append, close).
CSV	逗号分隔值	Dòuhào fēngé zhí	A file format where data is separated by commas, commonly used for tabular data.
JSON	JavaScript对象表示法	JavaScript duìxiàng biǎoshì fǎ	A lightweight data interchange format using human-readable text.
Exception Handling	异常处理	Yìcháng chǔlǐ	A mechanism to handle runtime errors gracefully (try-except, try-catch).
Validation	验证	Yànzhèng	Checking that data meets specific criteria before processing (range, type, presence, format).

English	中文	Pinyin	Definition
Verification	校验	Jiàoyàn	Checking that data has been entered correctly (double entry, visual check).

Unit 7: Algorithms

English	中文	Pinyin	Definition
Algorithm	算法	Suànfǎ	A step-by-step procedure for solving a problem or accomplishing a task.
Pseudocode	伪代码	Wěi dàimǎ	A high-level description of an algorithm using a mixture of natural language and programming constructs.
Flowchart	流程图	Liúchéng tú	A visual representation of an algorithm using standard symbols (start, process, decision, input/output).
Decomposition	分解	Fēnjiě	Breaking a complex problem into smaller, more manageable sub-problems.
Pattern Recognition	模式识别	Móshì shíbié	Identifying similarities, trends, or patterns in data or problems.
Abstraction	抽象	Chōuxiàng	Hiding unnecessary detail to focus on relevant features of a problem.
Linear Search	线性搜索	Xiànxìng sōusuǒ	A search algorithm that checks each element in turn until the target is found.
Binary Search	二分搜索	Èrfēn sōusuǒ	A search algorithm that repeatedly divides a sorted list in half to find the target.

English	中文	Pinyin	Definition
Bubble Sort	冒泡排序	Màopào páixù	A sorting algorithm that repeatedly swaps adjacent elements if they are in the wrong order.
Insertion Sort	插入排序	Chārù páixù	A sorting algorithm that builds a sorted list one element at a time by inserting each new element into its correct position.
Quicksort	快速排序	Kuàisù páixù	A divide-and-conquer sorting algorithm that selects a pivot and partitions the list around it.
Merge Sort	归并排序	Guībìng páixù	A divide-and-conquer sorting algorithm that splits the list, sorts the halves, and merges them.
Big-O Notation	大O符号	Dà O fúhào	A notation describing the upper bound of an algorithm's time or space complexity as input size grows.
Time Complexity	时间复杂度	Shíjiān fùzá dù	A measure of the computational time an algorithm takes relative to input size.
Space Complexity	空间复杂度	Kōngjiān fùzá dù	A measure of the memory an algorithm requires relative to input size.
☐ Karnaugh Map	卡诺图	Kǎ nuò tú	A graphical method for simplifying Boolean expressions by grouping adjacent cells.
☐ De Morgan's Laws	德摩根定律	Dé mógēn dìnglǜ	Boolean algebra rules for converting AND to OR (and vice versa) with negation: $\text{NOT}(A \text{ AND } B) = (\text{NOT } A) \text{ OR } (\text{NOT } B)$.

English	中文	Pinyin	Definition
☐ Truth Table	真值表	Zhēnzhí biǎo	A table showing all possible input combinations and their corresponding outputs for a logic circuit.
☐ Logic Gate	逻辑门	Luójí mén	A physical device implementing a Boolean function (AND, OR, NOT, NAND, NOR, XOR).
☐ Flip-Flop	触发器	Chùfā qì	A bistable circuit that stores one bit of data, used in registers and memory.
☐ Register	寄存器	Jìcún qì	A small, fast storage location within the CPU used to hold data temporarily.
☐ Counter	计数器	Jìshù qì	A sequential circuit that counts pulses and stores the current count.
Combinational Logic	组合逻辑	Zǔhé luójí	Logic circuits where the output depends only on the current inputs.
Sequential Logic	时序逻辑	Shíxù luójí	Logic circuits where the output depends on current inputs and previous states (memory).

Unit 8: Data Structures

English	中文	Pinyin	Definition
Stack	栈	Zhàn	A linear data structure following Last-In-First-Out (LIFO): push and pop operations.
Queue	队列	Duìliè	A linear data structure following First-In-First-Out (FIFO): enqueue and dequeue operations.

English	中文	Pinyin	Definition
Linked List	链表	Liànbǎo	A data structure where each element contains data and a pointer to the next element.
Pointer	指针	Zhǐzhēn	A variable that stores the memory address of another variable.
Node	节点	Jiédiǎn	An element in a linked list or tree containing data and references to other nodes.
Tree	树	Shù	A hierarchical data structure with a root node and child nodes.
Binary Tree	二叉树	ǐr chā shù	A tree where each node has at most two children (left and right).
Binary Search Tree (BST)	二叉搜索树	ǐr chā sōusuǒ shù	A binary tree where left children are smaller and right children are larger than their parent.
Traversal	遍历	Biànlì	The process of visiting all nodes in a tree or graph in a specific order (pre-order, in-order, post-order).
Graph	图	Tú	A data structure consisting of vertices (nodes) connected by edges.
Adjacency Matrix	邻接矩阵	Línjiē jǔzhèn	A 2D array representing a graph where cells indicate whether edges exist between vertices.
Adjacency List	邻接表	Línjiē biǎo	A representation of a graph where each vertex stores a list of its neighbouring vertices.
Hash Table	哈希表	Hāxī biǎo	A data structure that maps keys to values using a hash function for fast lookup.

English	中文	Pinyin	Definition
Hash Function	哈希函数	Hāxī hánshù	A function that converts a key into an index for storing data in a hash table.
Collision	冲突	Chōngtú	When two keys produce the same hash value, requiring a resolution strategy.
☐ Object-Oriented Programming (OOP)	面向对象编程	Miànxàng duìxiàng biānchéng	A programming paradigm based on objects that contain data (attributes) and code (methods).
☐ Class	类	Lèi	A blueprint for creating objects, defining attributes and methods.
☐ Object	对象	Duìxiàng	An instance of a class with its own data and behaviour.
☐ Encapsulation	封装	Fēngzhuāng	Hiding internal data and providing public methods to access it.
☐ Inheritance	继承	Jìchéng	A mechanism where a new class derives properties and methods from an existing class.
☐ Polymorphism	多态	Duōtài	The ability of different objects to respond to the same method call in different ways.
☐ Method Overriding	方法重写	Fāngfǎ chóngxiě	Redefining a method in a derived class that already exists in the base class.
☐ Getter	获取器	Huòqǔ qì	A method that returns the value of a private attribute.
☐ Setter	设置器	Shèzhì qì	A method that sets the value of a private attribute, often with validation.

English	中文	Pinyin	Definition
Artificial Intelligence (AI)	人工智能	Rén gōng zhì néng	The simulation of human intelligence processes by computer systems.
Machine Learning	机器学习	Jī qì xué xí	A subset of AI where systems learn from data without being explicitly programmed.
Supervised Learning	监督学习	Jiān dū xué xí	Training a model on labelled data (input-output pairs) to make predictions.
Unsupervised Learning	无监督学习	Wú jiān dū xué xí	Training a model on unlabelled data to find hidden patterns or structures.
Reinforcement Learning	强化学习	Qiáng huà xué xí	Training a model through trial and error, receiving rewards or penalties for actions.
☐ Neural Network	神经网络	Shén jīng wǎng luò	A computing system inspired by biological neurons, consisting of interconnected nodes organised in layers.
☐ Perceptron	感知器	Gǎn zhī qì	The simplest neural network unit, a binary classifier that makes decisions based on weighted inputs.
☐ Deep Learning	深度学习	Shēn dù xué xí	Machine learning using neural networks with many layers to model complex patterns.
☐ Backpropagation	反向传播	Fǎn xiàng chuán bō	An algorithm for training neural networks by adjusting weights based on the error of the output.
☐ Training Set	训练集	Xùn liàn jí	The data used to train a machine learning model.

English	中文	Pinyin	Definition
☐ Validation Set	验证集	Yànzhèng jí	The data used to tune model parameters and prevent overfitting during training.
☐ Test Set	测试集	Cèshì jí	The data used to evaluate the final performance of a trained model.
☐ Overfitting	过拟合	Guò nǐhé	When a model learns the training data too well, including noise, and performs poorly on new data.
☐ Underfitting	欠拟合	Qiàn nǐhé	When a model is too simple to capture the underlying pattern in the data.
☐ Feature Extraction	特征提取	Tèzhēng tíqǔ	The process of selecting or transforming raw data into features that improve model performance.
Natural Language Processing (NLP)	自然语言处理	Zìrán yǔyán chǔlǐ	AI techniques for understanding and generating human language.
Computer Vision	计算机视觉	Jìsuànjī shìjué	AI techniques for interpreting and understanding visual information from images or video.
Expert System	专家系统	Zhuānjiā xìtǒng	An AI system that emulates the decision-making ability of a human expert using rules and knowledge bases.
Knowledge Base	知识库	Zhīshì kù	A structured collection of facts and rules used by an expert system.

English	中文	Pinyin	Definition
Inference Engine	推理引擎	Tuīlǐ yǐnqíng	The component of an expert system that applies logical rules to the knowledge base to derive conclusions.
Robotics	机器人学	Jīqìrén xué	The branch of technology dealing with the design, construction, and operation of robots.
Sensor	传感器	Chuángǎn qì	A device that detects physical input (light, heat, motion) and converts it into electrical signals.
Actuator	执行器	Zhíxíng qì	A device that converts electrical signals into physical action (movement, sound).
Feedback Loop	反馈回路	Fǎnkuì huílù	A system where output is fed back as input to regulate behaviour (e.g., thermostat control).
Autonomous System	自主系统	Zìzhǔ xìtǒng	A system capable of operating independently without human intervention.
☐ Big Data	大数据	Dà shùjù	Extremely large datasets that require specialised tools and techniques to store, process, and analyse.
☐ Volume (Big Data)	体量	Tǐliàng	The sheer amount of data generated and stored.
☐ Velocity (Big Data)	速度	Sùdù	The speed at which data is generated and must be processed.
☐ Variety (Big Data)	多样性	Duōyàng xìng	The different types of data (structured, semi-structured, unstructured).
☐ Veracity (Big Data)	真实性	Zhēnshí xìng	The quality and accuracy of data.

English	中文	Pinyin	Definition
☐ Hadoop	Hadoop	Hadoop	An open-source framework for distributed storage and processing of big data across clusters.
☐ Spark	Spark	Spark	An open-source big data processing engine designed for speed and ease of use.

Unit 10: Boolean Logic & Advanced Theory

English	中文	Pinyin	Definition
Boolean Algebra	布尔代数	Bù'ěr dàishù	A branch of algebra dealing with binary variables and logical operations (AND, OR, NOT).
AND Gate	与门	Yǔ mén	A logic gate that outputs 1 only if all inputs are 1.
OR Gate	或门	Huò mén	A logic gate that outputs 1 if at least one input is 1.
NOT Gate	非门	Fēi mén	A logic gate that inverts the input: outputs 1 if input is 0, and vice versa.
NAND Gate	与非门	Yǔ fēi mén	A logic gate that outputs 0 only if all inputs are 1 (NOT AND).
NOR Gate	或非门	Huò fēi mén	A logic gate that outputs 1 only if all inputs are 0 (NOT OR).
XOR Gate	异或门	Yì huò mén	A logic gate that outputs 1 if the inputs are different.
Logic Circuit	逻辑电路	Luójí diànlù	A physical arrangement of logic gates that performs a Boolean function.
Half Adder	半加器	Bàn jiā qì	A circuit that adds two single bits, producing a sum and a carry.

English	中文	Pinyin	Definition
Full Adder	全加器	Quán jiā qì	A circuit that adds three bits (two inputs and a carry-in), producing a sum and a carry-out.
☐ Simplification	简化	Jiǎnhuà	The process of reducing a Boolean expression to its simplest form using algebraic rules or Karnaugh maps.
☐ Minimisation	最小化	Zuì xiǎo huà	Finding the simplest logic circuit that implements a given Boolean function, reducing cost and delay.
☐ Don't Care Condition	无关条件	Wúguān tiáojiàn	An input combination that will never occur, which can be treated as 0 or 1 to simplify the expression.

Additional A-Level Specific Terms

English	中文	Pinyin	Definition
☐ Systems Development Life Cycle (SDLC)	系统开发生命周期	Xìtǒng kāifā shēngmìng zhōuqī	The structured process of developing an information system: analysis, design, development, testing, implementation, maintenance.
☐ Feasibility Study	可行性研究	Kěxíngxìng yánjiū	An assessment of whether a proposed project is technically, economically, and operationally viable.
☐ Data Dictionary	数据字典	Shùjù zìdiǎn	A centralised repository of information about data: names, types, sizes, descriptions, and relationships.

English	中文	Pinyin	Definition
☐ Normalisation (Database)	数据库规范化	Shùjùkù guīfàn huà	The process of organising database tables to reduce redundancy and improve data integrity.
☐ First Normal Form (1NF)	第一范式	Dì yī fànshì	A database table with no repeating groups and a primary key defined.
☐ Second Normal Form (2NF)	第二范式	Dì èr fànshì	A table in 1NF where all non-key attributes are fully dependent on the primary key.
☐ Third Normal Form (3NF)	第三范式	Dì sān fànshì	A table in 2NF with no transitive dependencies (non-key attributes depend only on the key).
☐ Test Plan	测试计划	Cèshì jìhuà	A document describing the scope, approach, resources, and schedule of testing activities.
☐ Normal Data	正常数据	Zhèngcháng shùjù	Test data that is valid and within the expected range.
☐ Boundary Data	边界数据	Biānjiè shùjù	Test data at the extreme limits of valid ranges (e.g., minimum, maximum, just inside, just outside).
☐ Erroneous Data	错误数据	Cuòwù shùjù	Test data that is invalid or outside the expected range to test error handling.
☐ Black-box Testing	黑盒测试	Hēi hé cèshì	Testing without knowledge of internal code; focuses on inputs and outputs.
☐ White-box Testing	白盒测试	Bái hé cèshì	Testing with knowledge of internal code; focuses on path coverage and logic.

English	中文	Pinyin	Definition
☐ Technical Documentation	技术文档	Jìshù wéndàng	Documentation for developers or maintainers: system architecture, code comments, data structures.
☐ User Guide	用户指南	Yònghù zhǐnán	Documentation for end users explaining how to operate the system.
☐ Evaluation	评估	Pínggū	A critical review of the project against objectives, including strengths, limitations, and improvements.
☐ Maintenance	维护	Wéihù	The process of modifying a system after deployment to correct faults, improve performance, or adapt to changes.
☐ Iterative Development	迭代开发	Diédài kāifā	A development approach where the system is built in repeated cycles, refining with each iteration.
☐ Agile Methodology	敏捷方法	Mǐnjié fāngfǎ	A flexible approach to development emphasising collaboration, customer feedback, and rapid iteration.
☐ Version Control	版本控制	Bǎnběn kòngzhì	Tracking and managing changes to code over time (e.g., Git).
☐ Debugging	调试	Tiádòushì	The process of identifying, locating, and removing errors (bugs) from code.
☐ Trace Table	追踪表	Zhuīzōng biǎo	A table used to manually trace the values of variables through each step of an algorithm.

English	中文	Pinyin	Definition
☐ Stub	存根	Cúngēn	A simplified piece of code used to stand in for a more complex component during testing.
☐ Driver	驱动程序	Qūdòng chéngxù	A test harness that calls a module and passes test data to it.
☐ Computational Thinking	计算思维	Jìsuàn sīwéi	A problem-solving approach using decomposition, pattern recognition, abstraction, and algorithm design.
☐ Abstraction (Data)	数据抽象	Shùjù chōuxiàng	Simplifying complex data by modelling only the relevant attributes and behaviours.
☐ Heuristic	启发式	Qǐfā shì	A problem-solving approach that uses practical rules or educated guesses rather than guaranteed methods.
☐ Turing Test	图灵测试	Túlíng cèshì	A test of a machine's ability to exhibit intelligent behaviour indistinguishable from a human.
☐ Chinese Room Argument	中文房间论证	Zhōngwén fángjiān lùnzheng	A thought experiment by Searle arguing that syntax (symbol manipulation) is not sufficient for semantics (understanding).
☐ Ethics of AI	人工智能伦理	Rén gōng zhìnéng lúnlǐ	The moral principles governing the development and use of artificial intelligence systems.
☐ Bias in AI	人工智能偏见	Rén gōng zhìnéng piānjiàn	Systematic and unfair prejudice in AI systems, often stemming from biased training data.

English	中文	Pinyin	Definition
☐ Accountability	问责制	Wèn zé zhì	The obligation to explain and justify decisions, especially important for AI systems.
☐ Transparency	透明度	Tòumíng dù	The quality of being open and understandable, especially regarding how AI systems make decisions.
☐ Green Computing	绿色计算	Lǜsè jìsuàn	Environmentally sustainable computing practices to reduce energy consumption and e-waste.
☐ E-Waste	电子垃圾	Diànzǐ lājī	Discarded electronic devices and components, a growing environmental concern.
☐ Intellectual Property	知识产权	Zhīshì chǎnquán	Legal rights protecting creations of the mind, including software, patents, and copyrights.
☐ Open Source	开源	Kāi yuán	Software with source code available for anyone to view, modify, and distribute.
☐ Proprietary Software	专有软件	Zhuānyǒu ruǎnjiàn	Software owned by an individual or company, with restricted access to source code.
☐ GNU General Public License (GPL)	GNU通用公共许可证	GNU tōngyòng gōnggòng xǔkě zhèng	A widely used free software license ensuring users can run, study, share, and modify software.
☐ Computer Misuse Act	计算机滥用法	Jìsuànjī lànòng fǎ	UK legislation criminalising unauthorised access to computer systems and data.

English	中文	Pinyin	Definition
☐ Copyright	版权	Bǎnquán	Legal protection granting creators exclusive rights to reproduce, distribute, and display their work.
☐ Patent	专利	Zhuānlì	A legal right granting exclusive use of an invention for a limited period in exchange for public disclosure.
☐ Trademark	商标	Shāngbiāo	A recognisable sign, design, or expression identifying products or services of a particular source.
☐ Software License	软件许可证	Ruǎnjiàn xǔkě zhèng	A legal agreement defining how software can be used, distributed, and modified.
☐ End-User License Agreement (EULA)	最终用户许可协议	Zuìzhōng yònghù xǔkě xiéyì	A legal contract between software provider and user, often agreed to by clicking "I accept."
☐ Consequentialism	结果主义	Jiéguǒ zhǔyì	An ethical theory judging actions by their outcomes or consequences.
☐ Deontology	义务论	Yìwù lùn	An ethical theory judging actions by adherence to rules or duties, regardless of consequences.
☐ Virtue Ethics	美德伦理学	Měidé lúnlǐ xué	An ethical theory focusing on the character and virtues of the moral agent rather than rules or consequences.

How to Use This Glossary

For Students:

- ☐ Use the glossary to check understanding before and after each lesson.
- ☐ Create flashcards (English front, Chinese back) for spaced repetition.
- ☐ Practice writing definitions from memory, then check against the glossary.
- ☐ Use Pinyin to ensure correct pronunciation when discussing terms with peers or teachers.

For Teachers:

- ☐ Display key terms with Chinese translations at the start of each topic.
- ☐ Use bilingual terms in worksheets and assessments to support EAL students.
- ☐ Test students on term definitions in both English and Chinese.
- ☐ Encourage students to add their own terms to a personal glossary.

Resources

Cambridge International A-Level Computer Science (9618) Syllabus (2024 – 2026)

Cambridge IGCSE Computer Science (0478) Syllabus — for comparison of IGCSE vs. A-Level terms

Python 3.12 Documentation

VS Code Editor

VPA A2 Semester 2 Course Overview

VPA A2 Semester 2 Assessment Pack

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Note: Terms marked with ☐ are A-Level specific and not typically covered at IGCSE level.